

14, PDC analysis

Prepared the decoder for PDC with HODP as a reference. First, I checked the hit pattern and TDC response as a fellow. We can see two or three unexpected structures, especially X-plane. It would be a delayed induced signal, which has no more position information. We will eliminate the 2nd or 3rd pulse if it has the same wire. Next, check the ordering of directions.

After eliminating the unexpected information by the width threshold and time window, we can use simple clustering to remove accidental hits that are not members of each cluster. All the filtering was done on the base of the anaroot framework. We take the simple weighted average in the plane. Figure 20 shows the correlation between FDC1 and PDC. Here, the middle between PDC1 and PDC2 is used as a PDC position with deuteron beam RUN. Left- and right-hand sides show the correlation between FDC1 position and PDC position and angle, respectively. The upper and lower panels show the first or the second track of FDC1, respectively. Clearly, we can see a much stronger correlation in the second track of FDC1, which is consistent with our hypothesis; the second track is high-energy protons, as discussed above. One observable correlation between the vertical direction of the first track has a worse resolution than the second track. It suggests that the vertical angle of the second track by FDC1 has enough precision, which is the opposite conclusion discussed above in Fig. 13. We need to investigate the resolution evaluation more carefully.

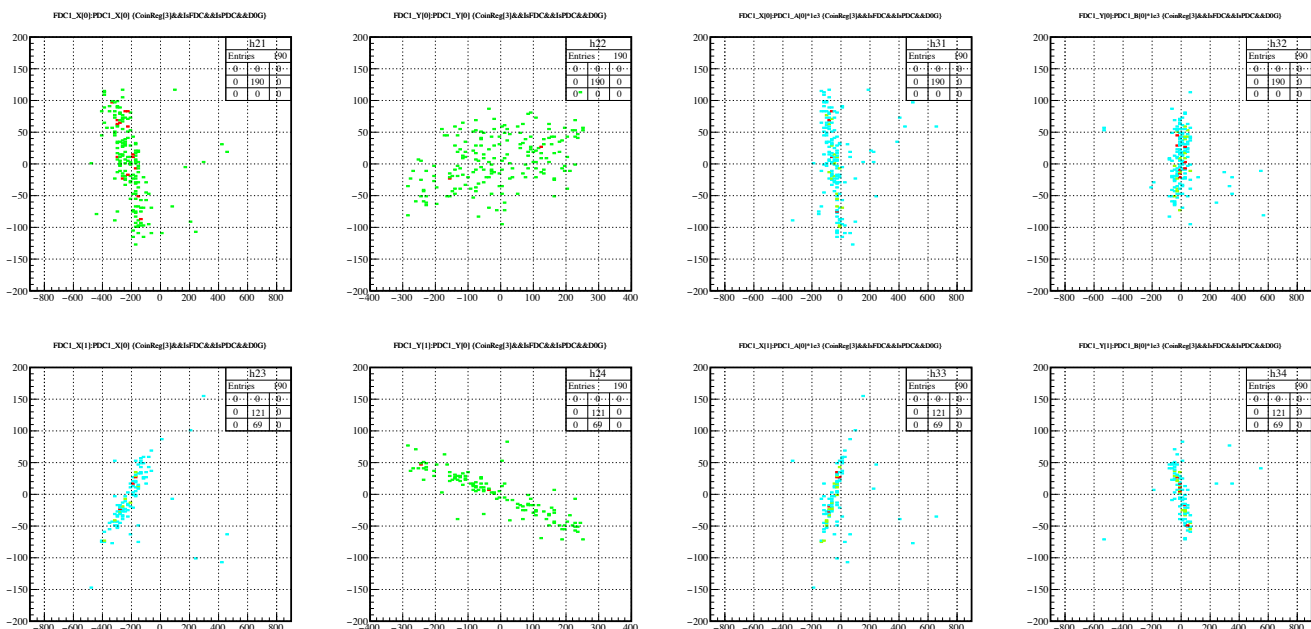


Figure 20, Correlation between FDC1 and PDC. Details in text.

To Do

- Develop the analysis code for PDC Including. Multi-Tracking capability
- Check coordinate and consistency with dimension (and photogrammetry) => Partly done.
- Clustering and position determination => almost coded.
- Efficiency and position resolution evaluations. => Not yet

>>**Notice**<<

According to the DB of PDC, I found a 1ch mismatch between X and 'U and V'. From geometry, we can check consistent using,

$$(U + V)/\sqrt{2} - X.$$

We can see a good correlation, but it has offset '-12 mm', which is exactly the same as a cathode step. The +12 mm is introduced by a PDC-X database.

>>Analysis Procedure<<

I coded the general procedure for the cathode readout chamber in the anaroot manner for PDC. First, restrict the time windows and width threshold. Here, we took a tight threshold of 60 ch. Second, I applied simple clustering and calculated the sum of the total width by the ToT. The more accurate position within the strip can be deduced from the charge balance of the top three channels.

Understanding the response function for the induced signal of PDC is very important. We have already known that the SECHS function;

$$f(x|a, \mu, \delta) = \frac{a}{\cosh^2(\pi(x - \mu)/\sigma)}$$

is a very good approximation. Here, the centroid μ from the strip center can be expressed by

$$\mu = \frac{\sigma}{\pi} \tanh^{-1} \left(\frac{\sqrt{Q_1/Q_3} - \sqrt{Q_1/Q_2}}{2 \sinh(\pi w/\sigma)} \right) \quad \sigma = \frac{\pi w}{2 \cosh^{-1}(\frac{1}{2}(\sqrt{Q_1/Q_3} + \sqrt{Q_1/Q_2}))}.$$

Here w is a strip size 12 mm for PDC.

More useful formula can be found on NIMA638(11)_134 for GANIL_MAYA (but typo) as;

$$\mu = \frac{w}{2} \frac{\log(1 + a_1) - \log(1 - a_1)}{\log(a_2 + \sqrt{a_2^2 - 1})}$$

$$a_1 = \frac{1}{2} \frac{\sqrt{Q_1/Q_3} - \sqrt{Q_1/Q_2}}{\sqrt{a_2^2 - 1}} \quad \text{and} \quad a_2 = \frac{1}{2} (\sqrt{Q_1/Q_3} + \sqrt{Q_1/Q_2})$$

The formula is the same as above.

In addition to SECHS, we prepared several response functions, such as **COG**, **COGW**, **Lorentzian**, and **Gaussian**, from reference NIMA366(95) 298.

COG(Center of Gravity :: weighted position)

$$x(= x_0 + \mu) = \frac{\sum_i q_i x_i}{\sum_i q_i}$$

COGW(Center of Gravity with bias),

$$x(= x_0 + \mu) = \frac{\sum_i q_i^n x_i}{\sum_i q_i^n}$$

Lorentzian,

$$\mu = \frac{w}{2} \frac{Q_1 Q_2 - Q_1 Q_3}{Q_1 Q_3 + Q_1 Q_2 - 2 Q_2 Q_3}$$

Gaussian,

$$\mu = \frac{w}{2} \frac{\log(Q_1/Q_3) - \log(Q_1/Q_2)}{\log(Q_1/Q_3) + \log(Q_1/Q_2)}$$

A possible problem in this procedure is the non-linearity of ToT information, which is difficult to estimate. An alternative method could be applied to adjust the phenomenological determination of position from charge differences between neighboring channels, which was used for SRPPAC at CNS. We are investigating the best way to deduce the position with the PDC.

>>Coordinate (of PDC)<<

I trust the coordinates for PDC defined by the DB, which show that PDC1 is downstream and PDC2 is upstream. The distance between is 616 mm [dimension is 600mm]. This is confirmed by the tracking of the HODP, which is located around 400 mm upstream from PDC2. Since we use the right-handed coordinate usually, the z-direction is opposite from the beam. The horizontal and vertical direction is a low momentum side and upper side, respectively. The vertical and U-V coordinates of PDC were confirmed by the HODP time difference.

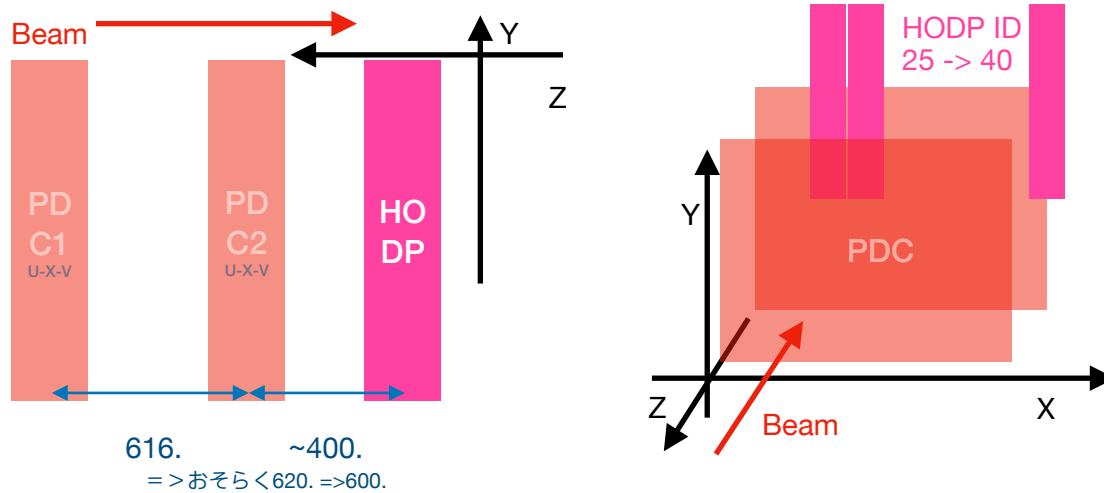


Figure 21, Schematic view of coordinates about PDCs and HODP.

>>TDC and TOT distribution <<

Figure 22 shows the TDC distribution and its correlation with the ToT. At the moment, I choose only the first component of the TDC distribution indicated by the arrows in the figure and a width higher than 60 ch as a normal hit. Only the X plane of both PDCs has larger multiple components, which would be due to higher gain than 'U' and 'V.' Here, I ignore these components by filtering the timing.

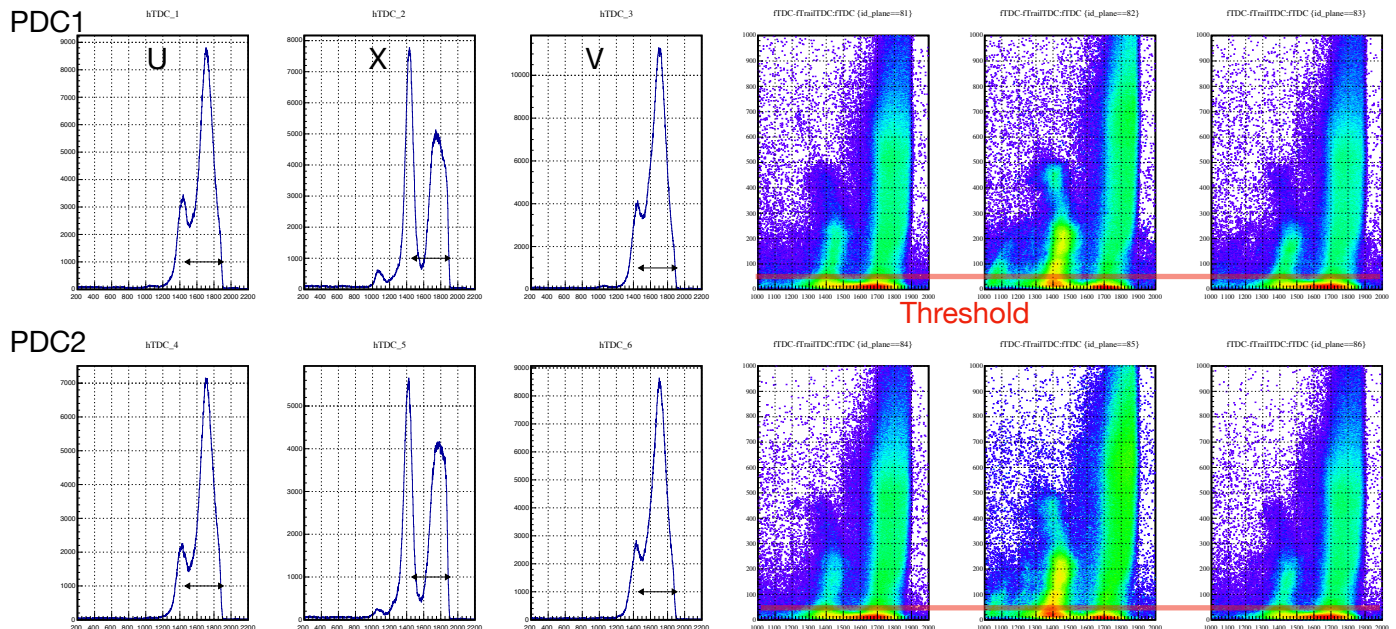


Figure 22, TDC and its correlation with ToT distribution of PDC 6 planes with 'D' beam #290.

>>Width analysis<<

Figure 23 shows the cluster size distribution after filtering for 'D' beam #290 and #ALL but only stopped deuteron as gated by the NINJA. Here, the cluster size is defined by the number of hits following neighboring channels. I did not take into account that the two clusters are close to each other (typically within 5ch = 60 mm). At least two hits are required as a cluster. Typically, one cluster consists of 4 or 5 hits. If we select the high energy proton from d+p elastic scattering at the forward angle, the cluster size becomes lower but still 3-4 hits. Although the present width threshold would be low enough, it still remains a space.

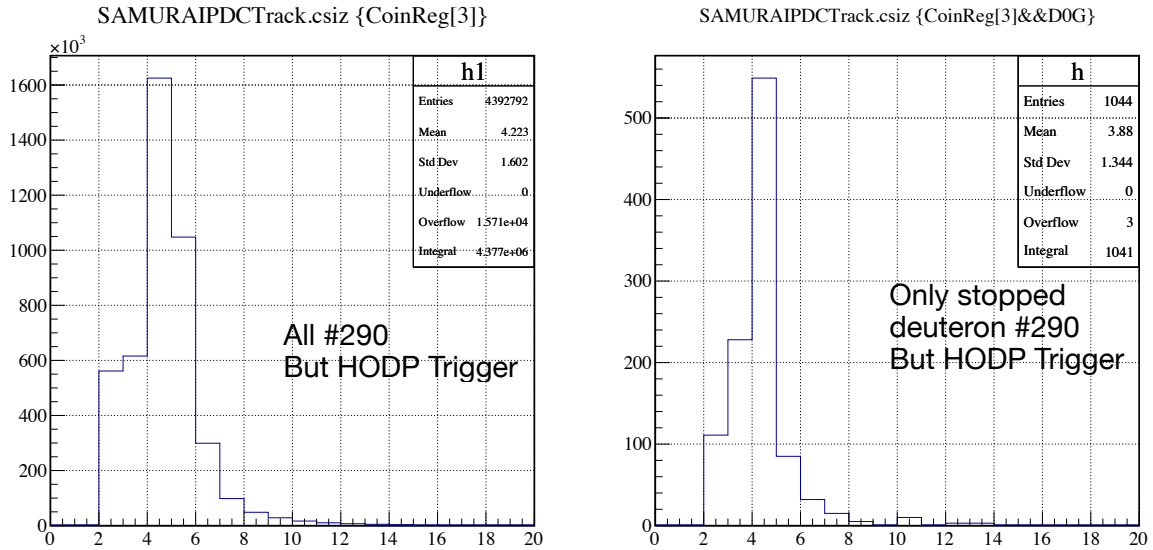


Figure 23, Cluster size distribution of PDC /place with 'D' beam #290(Left) and #ALL with high energy protons(Right).

Figure 24 shows the correlation between cluster size and averaged ToT distribution per hit(channel). The peak positions of C.Size = 3, 4, and 5 are similar. On the other hand, C.Size = 2 is smaller and C.Size >5. This means that a higher cluster size is a candidate for two clusters that are spatially close to each other. C.Size = 2 is missed position information compared to the higher size cases, which could be better to reduce the threshold.

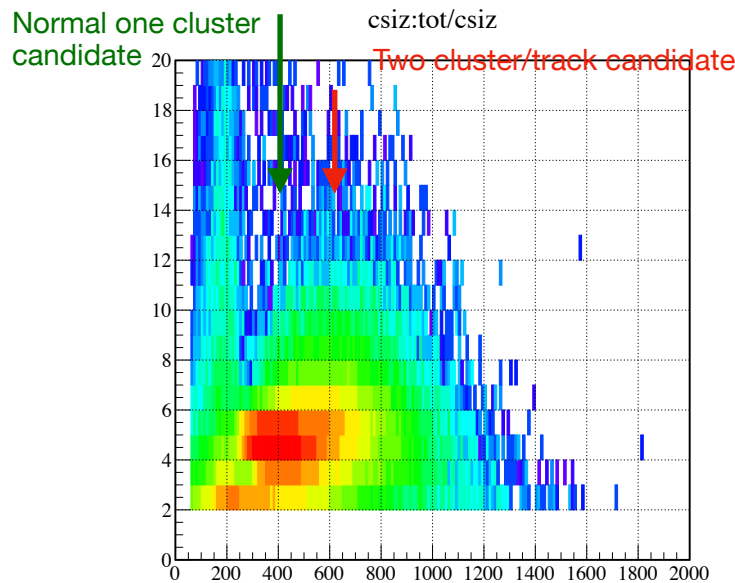


Figure 24, Cluster size and averaged ToT distribution.

Figure 25 shows the comparison between PDC-tot and HODP-Q. Since the energies of all the particles are high enough to penetrate the HODP, a clear linear correlation was observed as expected. We can see several groups corresponding to H/He/Li and only see H/He with 'Be' #310 and 'D' #290 beam, respectively. A bit of visible deviation from the black can be seen for the proton region because only the proton has a much faster velocity here after the SAMURAI magnet.

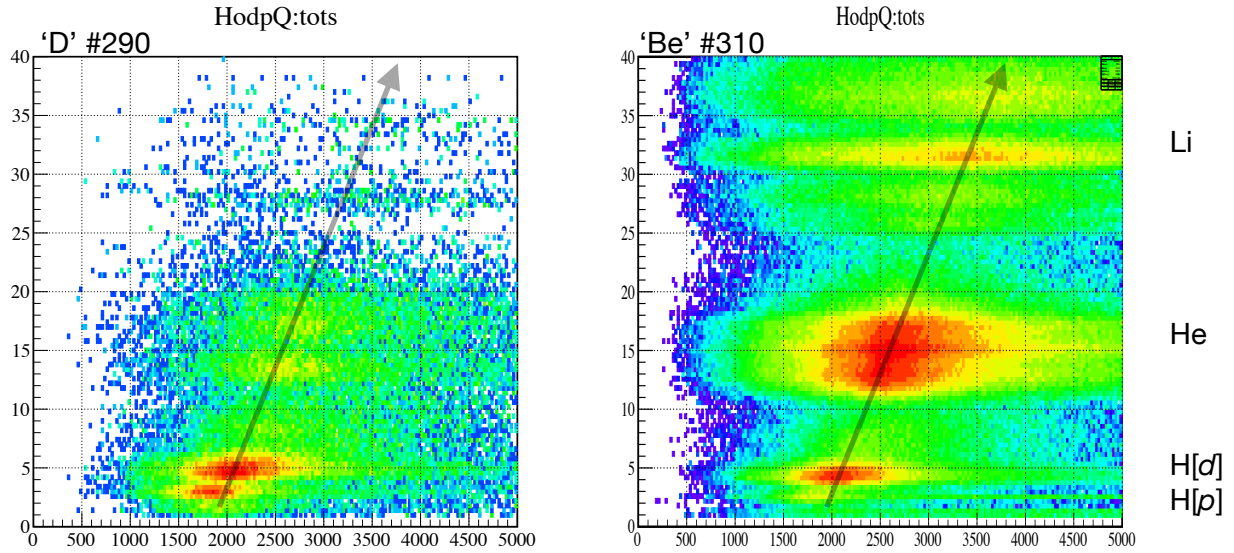


Figure 25, Correlation between ToT of PDC and HODP Q information with 'D' beam #290(Left) and 'Be' beam #310(Right)

>>Short summary about PDC<<

We have developed the analysis code for PDC in an anaroot manner and checked several correlations with other detectors. It works well, generally speaking.

We perfectly confirm the coordinates of PDCs, which would be a nice reference to determine the global coordinates of other detectors: FDC1, BDC, and NINJA.

We can see a reasonable correlation between FDC1 positions and PDC positions and angles, HODP-ID and -Q by extrapolated position and the averaged ToT of PDC.

At the moment, two clusters close to each other are treated as one cluster, which destroys the capability of multi-charged particle detection with a small relative energy. We have to investigate these issues for our physics goal.

Further optimization for several parameters will be investigated for better efficiency and resolution. At present, efficiency is good enough. It looks comparable to the first hit with the FDC1. But position resolution looks worse than expected, around 1-2 mm in σ per plane.

Now, we are thinking of treating two clusters but very close. First, we can set modeling by the simplified Gaussian distribution as

$$Q_{i-1} = Ae^{-\frac{(x_i - (x_0 - W))^2}{2\sigma^2}}, Q_i = Ae^{-\frac{(x_i - x_0)^2}{2\sigma^2}}, \text{ and } Q_{i+1} = Ae^{-\frac{(x_i - (x_0 + W))^2}{2\sigma^2}}$$

$$\sigma^2 = -W^2 / \log(Q_{i-1}Q_{i+1}/Q_i^2), \text{ and } \sqrt{\frac{Q_{i-1}}{Q_{i+1}}} = e^{\frac{W(x_i - x_0)}{\sigma^2}}.$$

$$Q_{i-n} = Q_i^{1-n^2} \times Q_{i-1}^{\frac{n(n+1)}{2}} \times Q_{i+1}^{\frac{n(n-1)}{2}} = \sqrt{\frac{Q_{i-1}^{n(n+1)} Q_{i+1}^{n(n-1)}}{Q_i^{n^2-1}}}$$

$$Q_{i-2} = \frac{Q_{i-1}^3 Q_{i+1}}{Q_i^3} = Q_{i+1} \left(\frac{Q_{i-1}}{Q_i} \right)^3$$

$$Q_{i-3} = \frac{Q_{i-1}^6 Q_{i+1}^3}{Q_i^8} = Q_{i+1} \left(\frac{Q_{i-1}}{Q_i} \right)^6 \left(\frac{Q_{i+1}}{Q_i} \right)^2$$

$$Q_{i-4} = \frac{Q_{i-1}^{10} Q_{i+1}^6}{Q_i^{15}} = Q_{i+1} \left(\frac{Q_{i-1}}{Q_i} \right)^{10} \left(\frac{Q_{i+1}}{Q_i} \right)^5$$

$$Q_{i-5} = \frac{Q_{i-1}^{15} Q_{i+1}^{10}}{Q_i^{24}} = Q_{i+1} \left(\frac{Q_{i-1}}{Q_i} \right)^{15} \left(\frac{Q_{i+1}}{Q_i} \right)^9$$

つまりGaussianを仮定したDecomposeが可能か? Now we assume that i and j are hit position, $|i-j| \geq 3$, and only thinking $i+2$ cases

$$Q_{i+1} = Q_{i+1}^* - Q_{j+1} Q_j^{-3} (Q_{i+2}^* - Q_{i-1} Q_i^{-3} Q_{i+1}^3)^3$$

and

$$Q_{j-1} = Q_{i+2}^* - Q_{i-1} Q_i^{-3} (Q_{i+1}^* - Q_{j+1} Q_j^{-3} Q_{j-1}^3)^3$$

$i+1$ および $j-1$ の9次方程式になっていて簡単に解けない。iterativeに解くのが現実的か

解析するのは3ズレから

補正を入れるのは3、4、5ズレまで

3、は Q_{i-2} 及び Q_{i-3}

4、5は Q_{i-3} を解析する必要がある



これは諦める

これは解析する。

つまり最近接距離は12mm x 3strip = 36mmくらいが限界。

Eye-Scanをする限りストリップ抜けが多い

なんで? 局所的にGainが低い?

小林さんに問い合わせ (2024.07.30)

Xは両方のInducedがあるのと直交してないのでDrift補正しないとまともな値にはならない
=> あとで真面目に考える

現状はTOT-TDCの相関でU,Vの分解能をだしている。この方法はU, Vのみ可能

CRM(3hit)での到達分解能は $350 \mu\text{m}[\sigma]$ 以下, DW(2hit)で $500 \mu\text{m}[\sigma]$ 以下

CRMのEffectiveGapは現象論。実験ごとにことなる。[NonLinearityの現象論的補正込み?]

Gain調整も行っている。10%程度の補正。

ということらしい

今は3hitを使うSECHS_Function。同じ処方箋で分解能を見てみると1.2-1.5 mm $[\sigma]$ 。Driftは詳細解析なし、Gain調整なしと思えばそれほど悪くないかも。どうやってモニタするかを考える。

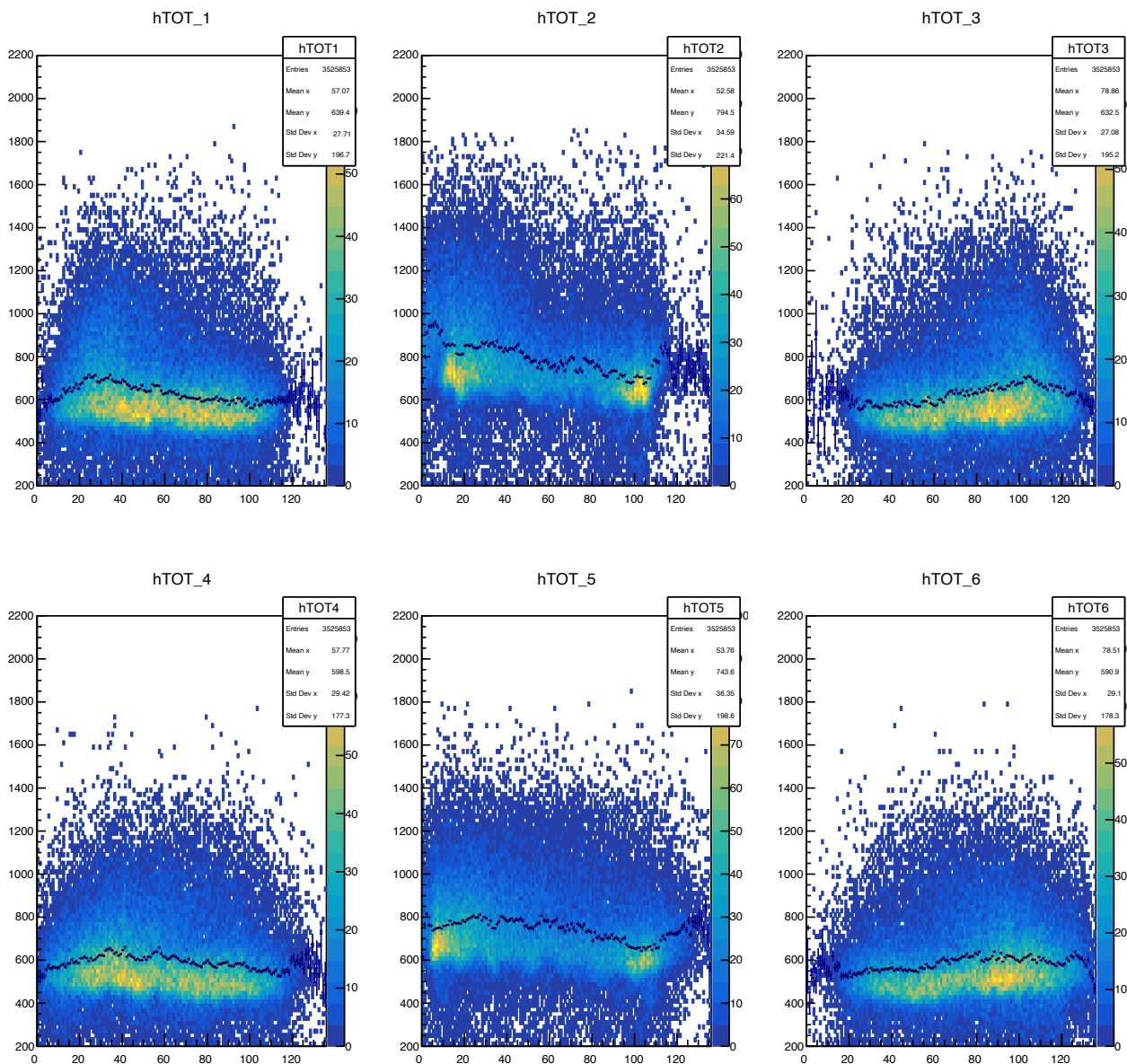


Figure 26, Tot-stripID plot of all the PDC planes. #290.

Figure 26 shows the Tot strip-ID plot of the PDCs. The maximum ToT defines the Tot in each event. The ProfileX function of ROOT was applied and plotted in the Figure.

数チャンネル補正した状態でTot Chを見てみた。合わせてProfile計算したものをPlotしている。RUNは#290. とりあえず滑らかになるように1-5%くらいで補正していく。ParameterにToT Gain追加、解析にGainを埋め込んだ。Clusteringする時のWidthで入れ込んでいる。Clusps内のwidthは補正されたもの。

Cathode解析とDrift解析の相関(#0290)。実際は対応するTDCとの相関を見る。間隔16mm周期で相関が見える。ただし、現在違うArrayの相関を見ているのでゴミが多く残っているはず。統計のためかBeビームの方が見えやすいので、#0311を解析。UVは思った通り、Z位置にはそれほどSensitiveではない。1mmは有意にでてるように見えるが、二本がずれているようにも見える。まだ努力が必要。

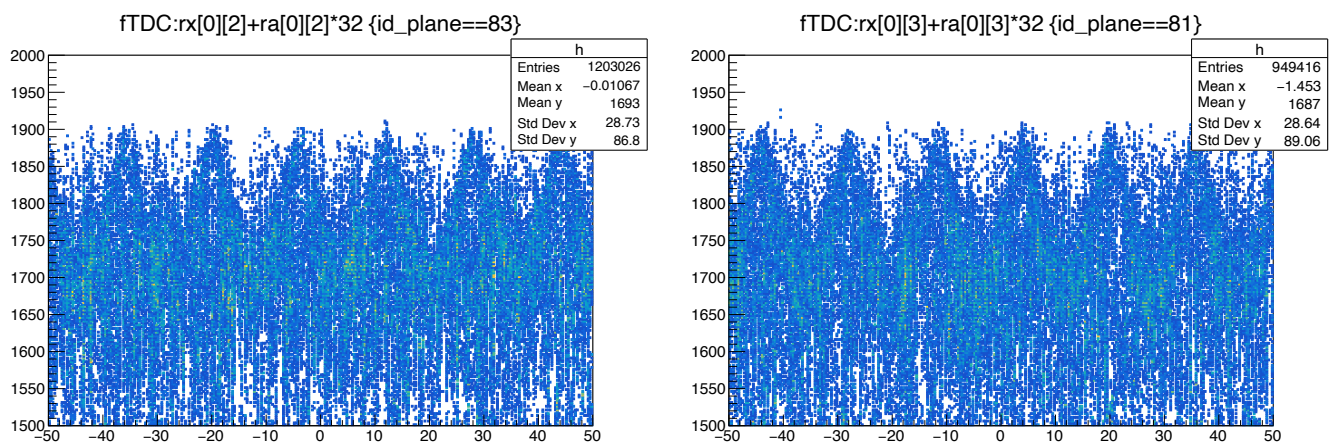


Figure 27, TDC - Position by Charge analysis correlation on #0290

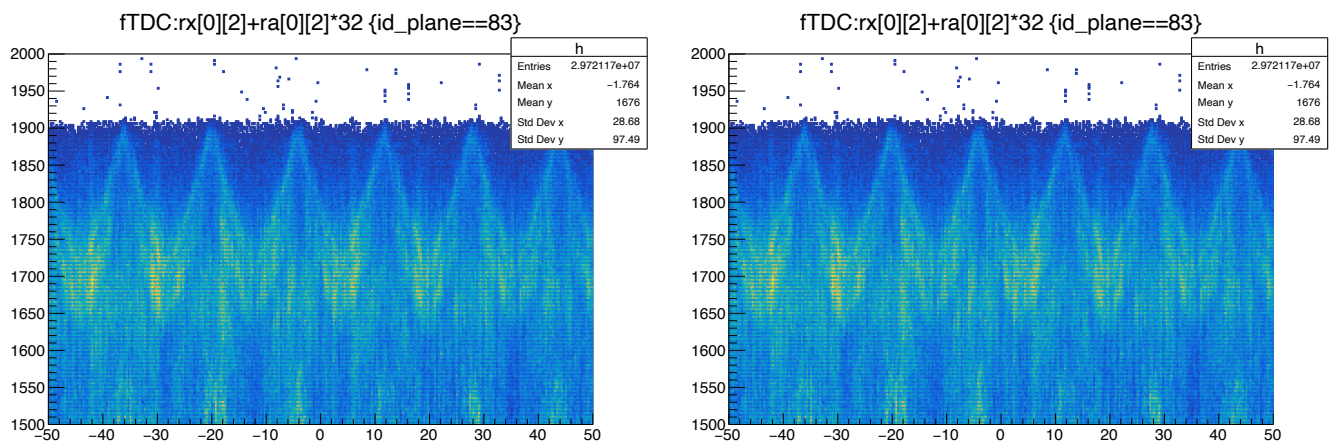


Figure 28, Same as Fig.27, but #0311

First, the QDC gains of HodoP are adjusted as follows: Figure 29 shows the PDC horizontal position and adjusted QDC of HodoP on the #0290(D) and #0310(Be). Except for the higher Q for the low momentum side, the gains of all bars were adjusted.

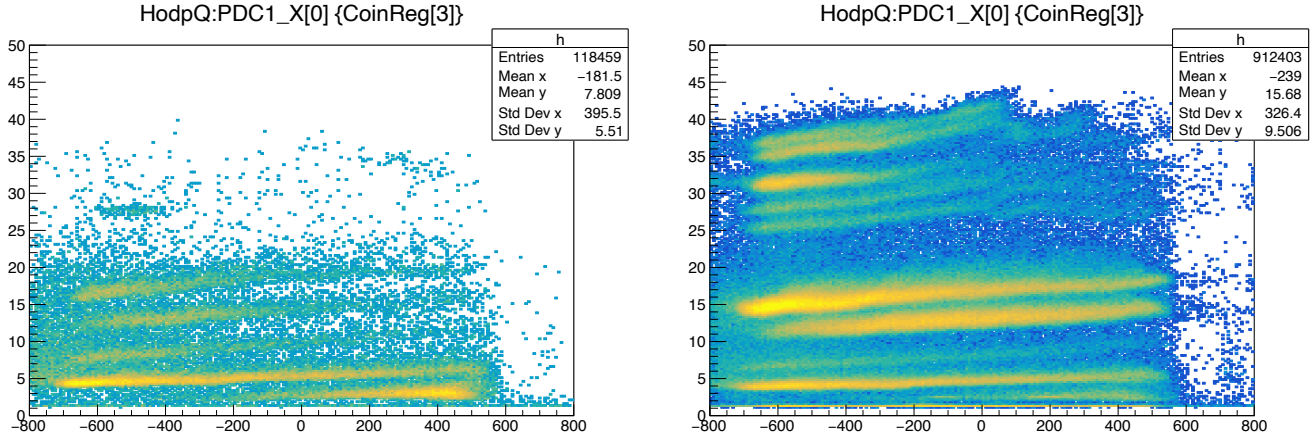


Figure 29, Q and PDC position correlation on #0290(Left) and #0310(Right)

Figure 30 shows the ToF and dE correlation with the HODP trigger. We can clearly see several foci, but they have unexpected tails. Here, ToF is a time difference from the trigger by SBT. They would come from the finite momentum broadening with the SAMURAI spectrometer, which simply corrected by introducing horizontal positions on the HodoP for a candidate. For hydrogens, it would be different.

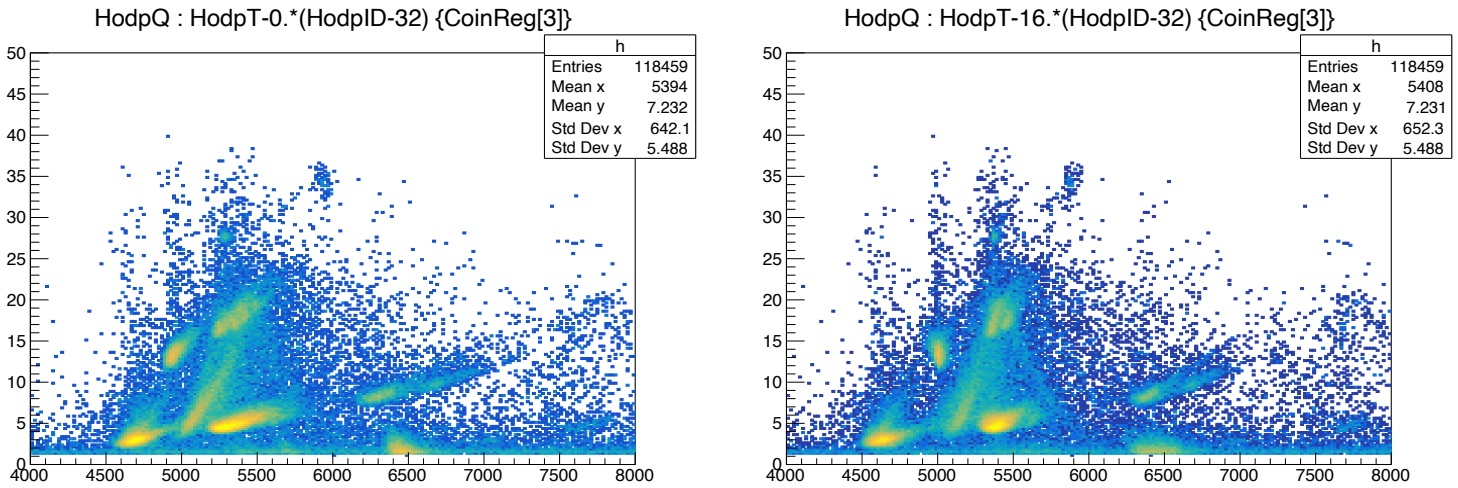


Figure 30, ToF-dE of HodoP plots without(Left) and with(Right) correction by position #290.

The observed foci can be speculated as H/D/T and 3He/4He, but there is an unexpected focus around 5100 ch and 6500 ch. To investigate these events, the PDC position-angle correlation was calculated in Fig. 31. Figure 31 shows the position-angle correlation by the PDC tracking on the data #0290 and #0310. Very narrow liens can be seen on both RUNs but in slightly different positions. They indicate the reaction particle between the beam and the exit flange of the SAMURAI chamber. Since the beam condition and the magnetic field are different, the lines have the position information hit on the beam directly. With the selection of foci in Fig.31, the clear correlation from H/D/T and slight α can be seen in Fig. 32. In Be RUN, several particles have still remained in the selection. The second focus in Fig.30 is naturally connected to the D focus in Fig. 31. It would be the low energy deuteron from the beam after the interaction but no proper tracking

information by the PDCs. The focus can not be seen on the Be RUN. It supports these events from the deuteron beam itself but with very low energy.

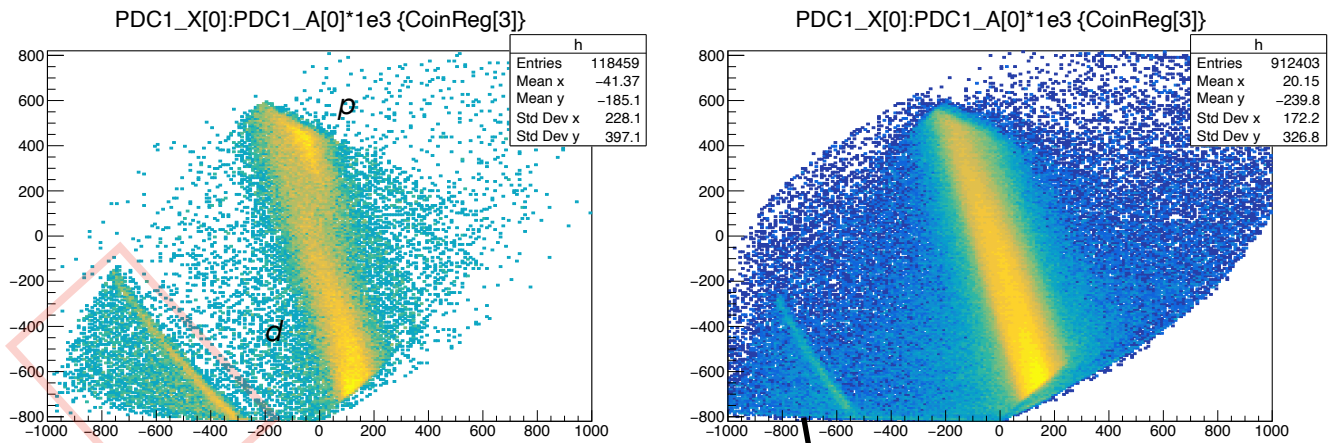


Figure 31, PDC position-angle correlations on #0290(Left) and #0310(Right).

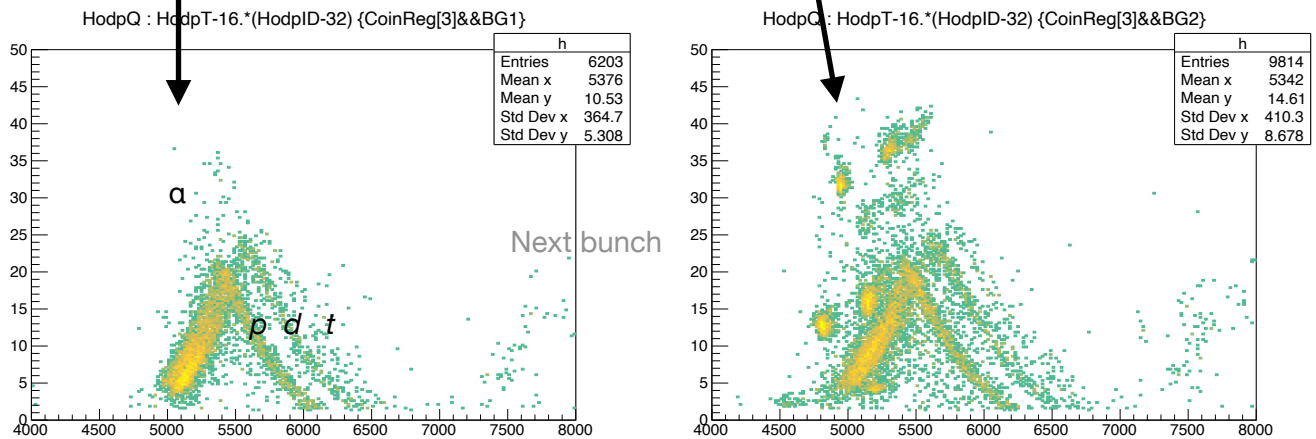


Figure 32, ToF-dE of HodoP plots with correction under the condition of foci on #0290 and #0310.

To confirm the PID by the HodoP, a similar analysis was done on #0310 in Fig.33. A clear line of the same rigidity particles by D/⁴He/⁶Li and the absence of ⁸Be (and ⁵He) were clearly identified.

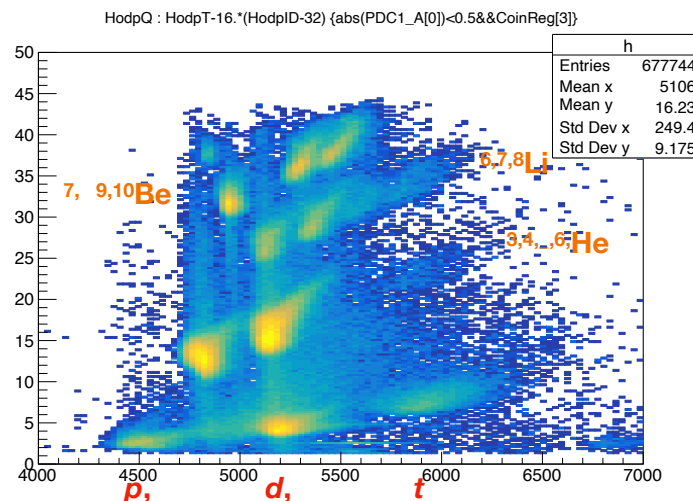


Figure 33, same as 30 with the correction on #0310.